TECHNICAL REFERENCE · PRODUCTION PRIORITY BOARD

PostgreSQL Data Model

All PostgreSQL tables and views directly used by the application, including relationships, join keys, field usage, transformation rules, and operational constraints.

Scope
Current application runtime

Prepared
18 September 2026

Source
Azure PostgreSQL / D365FO staging schema

Inventory
6 tables · 2 views

01 · SCOPE

What this model covers

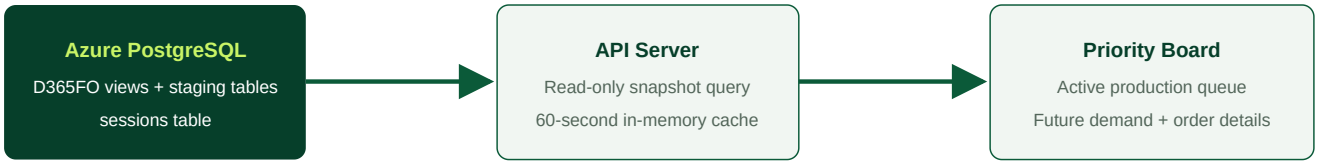
INCLUDED

- Every PostgreSQL object directly referenced by the API or session store.
- The columns actually read by the application.
- Logical relationships and normalized join keys.
- Derived dates, enrichment rules, and linkage validation.
- Live approximate row counts for physical tables.

NOT INCLUDED

- Underlying base objects hidden inside the two database views.
- Columns present in source tables but never read by this application.
- Admin Console data, which is accessed through an HTTP API rather than PostgreSQL.
- Browser-only state and generated API response types.

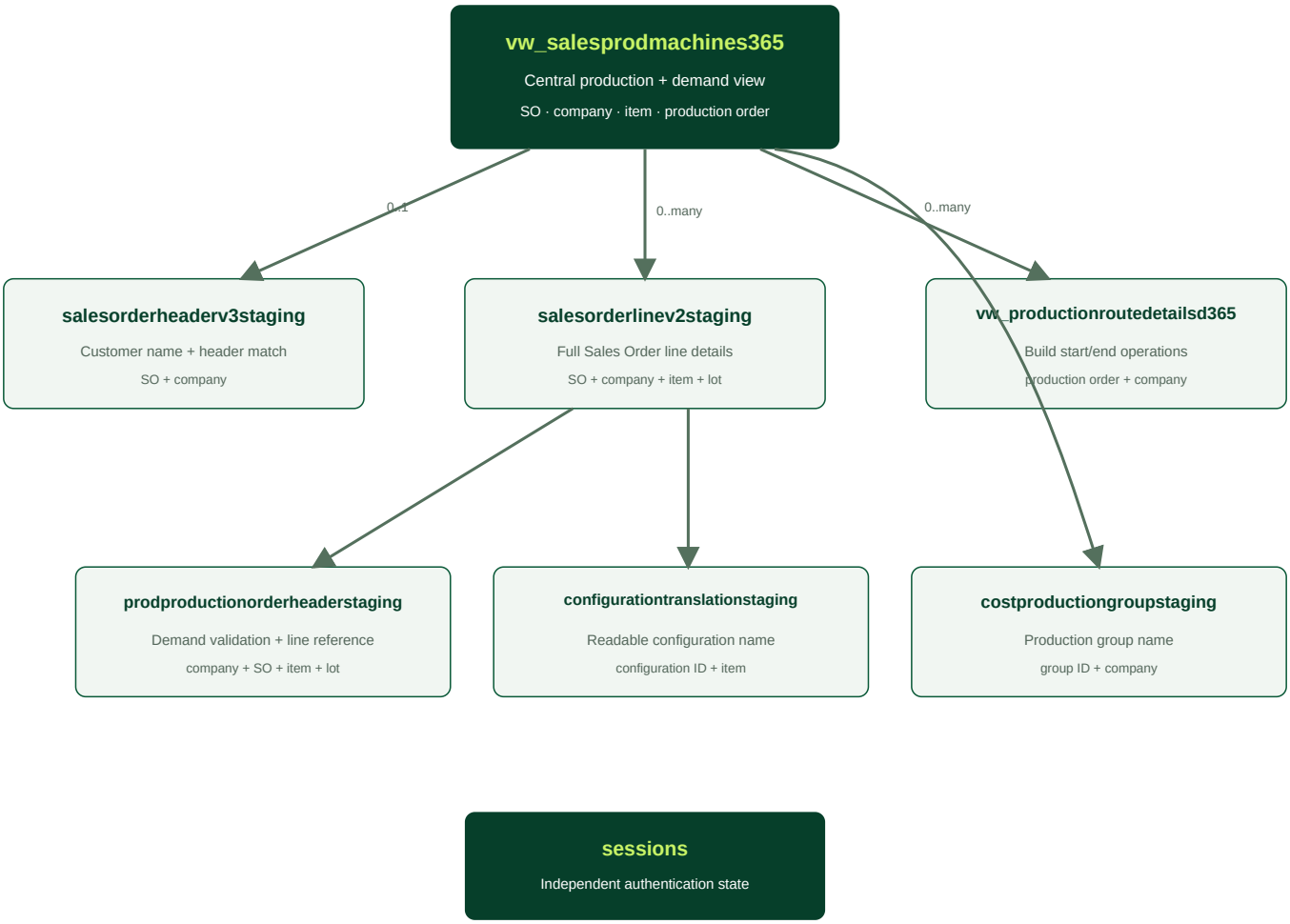
RUNTIME ARCHITECTURE



The browser never connects directly to PostgreSQL. Database credentials and source access remain server-side.

02 · ENTITY RELATIONSHIPS

Logical data model



KEY CONVENTION

Most D365FO relationships are company-aware. dataareaid must be included with the business identifier to prevent cross-company matches.

PostgreSQL objects used

Object	Type	Approx. rows	Primary purpose
vw_salesprodmachines365	View	View	Primary production-priority source. One working set combines sales demand, production-order status, classifications, dates, resource routing, and descriptive fields.
salesorderheaderv3staging	Table	≈112,743	Sales Order header enrichment and linkage validation.
salesorderlinev2staging	Table	≈355,646	Provides the complete Sales Order line list and connects sales demand to production demand.
prodproductionorderheaderstaging	Table	≈172,520	Validates the production-demand relationship and resolves a production-order reference for each Sales Order line.
vw_productionroutedetailsd365	View	View	Supplies dated manufacturing route operations used to derive build start and end dates.
ecoresproductmasterconfigurationtranslationstaging	Table	≈8,524	Translates a product configuration code into its readable configuration name.
costproductiongroupstaging	Table	≈96	Resolves a production-group code to its readable production-group name.
sessions	Table	≈42	Server-side Express session storage for authenticated Production Priority Board users.

RELATIONSHIP SUMMARY

From	To	Join key	Cardinality
Machine view	Sales Order header	`salesordernumber + dataareaid`	0..1
Machine view	Sales Order lines	`salesordernumber + dataareaid`	0..many
Sales Order header	Sales Order lines	`salesordernumber + dataareaid`	1..many
Sales Order lines	Production header	`company + Sales Order + item + inventory lot`	0..many
Machine view	Route details	`productionordernumber + dataareaid`	0..many
Sales Order lines	Configuration translation	`configuration ID + item number`	0..1
Machine view	Production group	`productiongroupid + dataareaid`	0..1

Physical-table row counts are PostgreSQL statistics captured on 18 September 2026 and are approximate. Views do not expose a stable stored row count.

OBJECT 01 · VIEW

vw_salesprodmachines365

View

ESTIMATED ROWS

Primary production-priority source. One working set combines sales demand, production-order status, classifications, dates, resource routing, and descriptive fields.

JOIN / IDENTITY FIELDS

- salesordernumber
- dataareaid
- itemnumber
- productionordernumber
- productiongroupid

APPLICATION OUTPUT

Board rows, future demand, customer/destination fallbacks, work-order and sales-order identifiers, dates, status, resource/stage, classification filters, description, quantity, and notes.

COLUMNS READ BY THE APPLICATION

salesordernumber	dataareaid	salesorderpoolid
salesordername	deliveryaddressname	confirmedshippingdate
requestedshippingdate	ordercreationdatetime	engineeringnotes
itemnumber	itemdesc	orderedsalesquantity
deliverydate	scheduledstartdate	productionorderstatus
starteddate	endeddate	status
productionordername	productionordernumber	sc1
sc2	sc3	name
Resource1	Resource	Task
Assy_Resource	MachineResource	productiongroupid

IMPLEMENTATION NOTE

Filtered first into a MATERIALIZED working set. The API adds a synthetic audit_row_id so all later enrichments aggregate back to the correct source row.

OBJECT 02 · TABLE

salesorderheaderv3staging

≈112,743
ESTIMATED ROWS

Sales Order header enrichment and linkage validation.

JOIN / IDENTITY FIELDS

- salesordernumber
- dataareaid

APPLICATION OUTPUT

Authoritative Customer Name and evidence that the Sales Order header exists.

COLUMNS READ BY THE APPLICATION

salesordernumber	dataareaid	salesordername

IMPLEMENTATION NOTE

Customer Name comes from salesordername. deliveryaddressname is not a customer name; it remains a ship-to/destination label.

OBJECT 03 · TABLE

salesorderlinev2staging

≈355,646
ESTIMATED ROWS

Provides the complete Sales Order line list and connects sales demand to production demand.

JOIN / IDENTITY FIELDS

- salesordernumber
- dataareaid
- itemnumber
- inventorylotid

APPLICATION OUTPUT

Line number, item, description, configuration, quantity, unit, status, inventory-lot demand ID, and Sales Order line linkage status.

COLUMNS READ BY THE APPLICATION

salesordernumber	dataareaid	linenum
itemnumber	itemdesc	linedescription
productconfigurationid	orderedsalesquantity	salesunitsymbol
salesorderlinestatus	inventorylotid	

IMPLEMENTATION NOTE

The detail panel intentionally loads every line for the Sales Order, not only the line matching the board row’s item.

OBJECT 04 · TABLE

prodproductionorderheaderstaging

≈172,520
ESTIMATED ROWS

Validates the production-demand relationship and resolves a production-order reference for each Sales Order line.

JOIN / IDENTITY FIELDS

- `productionordernumber`
- `demandsalesordernumber`
- `dataareaid`
- `itemnumber`
- `demandsalesorderlineinventorylotid`

APPLICATION OUTPUT

Production-demand linkage status and the line-level production-order Reference shown in order details.

COLUMNS READ BY THE APPLICATION

`productionordernumber`	`demandsalesordernumber`	`dataareaid`
`itemnumber`	`demandsalesorderlineinventorylotid`	

IMPLEMENTATION NOTE

Reference lookup is deduplicated once in a MATERIALIZED CTE, keyed by company, Sales Order, item, and demand lot.

OBJECT 05 · VIEW

vw_productionroutedetailsd365

View

ESTIMATED ROWS

Supplies dated manufacturing route operations used to derive build start and end dates.

JOIN / IDENTITY FIELDS

- productionordernumber
- dataareaid
- operationnumber

APPLICATION OUTPUT

Build Start Date from the first production operation and Build End Date from the last production operation.

COLUMNS READ BY THE APPLICATION

productionordernumber	dataareaid	operationnumber
operationname	scheduledfromdate	scheduledenddate

IMPLEMENTATION NOTE

Warehouse Pick and Warehouse Receive operations are excluded. Placeholder dates before year 2000 are ignored.

OBJECT 06 · TABLE

ecoresproductmasterconfigurationtranslationstaging

≈8,524
ESTIMATED ROWS

Translates a product configuration code into its readable configuration name.

JOIN / IDENTITY FIELDS

- `productmasterconfigurationid`
- `productmasternumber`
- `languageid`

APPLICATION OUTPUT

Configuration Name in the Sales Order line details.

COLUMNS READ BY THE APPLICATION

- `productmasterconfigurationid`
- `productmasternumber`
- `translatedconfigurationname`
- `languageid`

IMPLEMENTATION NOTE

Rows are deduplicated by configuration and item; en-us is preferred when multiple languages exist.

OBJECT 07 · TABLE

costproductiongroupstaging

≈96
ESTIMATED ROWS

Resolves a production-group code to its readable production-group name.

JOIN / IDENTITY FIELDS

- groupid
- dataareaid

APPLICATION OUTPUT

Production Group Name used in the board and detail context.

COLUMNS READ BY THE APPLICATION

- groupid
- dataareaid
- groupname
- tomodifieddatetime

IMPLEMENTATION NOTE

The newest row by tomodifieddatetime is selected for each company and group ID.

OBJECT 08 · TABLE

sessions

≈42

ESTIMATED ROWS

Server-side Express session storage for authenticated Production Priority Board users.

JOIN / IDENTITY FIELDS

`sid`

APPLICATION OUTPUT

Persistent authenticated session state and expiration control.

COLUMNS READ BY THE APPLICATION

`sid`	`sess`	`expire`

IMPLEMENTATION NOTE

Independent of production data. sid is the primary key; expire is indexed for cleanup. sess is JSONB.

How the model becomes board data

1. Establish the working set

Filter the central machine view to required production-priority rows, assign audit_row_id, and materialize the result.

2. Validate linkage

Join Sales Order header, matching Sales Order line, and production-demand header. Aggregate header/line/production match flags per source row.

3. Derive build dates

Join route operations by production order and company. Exclude warehouse operations and invalid pre-2000 dates; choose first and last valid production operations.

4. Build complete order-line details

Load all lines for each Sales Order. Enrich configuration names and line-level production references through set-based, deduplicated lookups.

5. Enrich production group

Resolve the latest company-specific group name from the production-group staging table.

6. Normalize API output

Split rows into active orders and future demand, normalize dates/statuses, calculate exception flags, and return a read-only snapshot.

Data quality and performance rules

Normalize string keys

Join keys use BTRIM and empty-string handling because D365FO staging values can contain whitespace or blanks.

Always retain company context

Use dataareaid with Sales Order, production order, and production-group identifiers whenever the source provides it.

Do not confuse customer and destination

Customer Name is salesorderheaderv3staging.salesordername. deliveryaddressname is only the ship-to destination.

Avoid correlated large-table lookups

Sales lines and production headers are large. Deduplicate once in MATERIALIZED CTEs and use set-based joins; do not use a per-row LATERAL scan.

Treat dates before 2000 as placeholders

Route sources can carry 1900-era placeholder dates. They are excluded before selecting build dates.

Views are contracts

This document models the columns the application consumes from each view. Changes inside a view can affect semantics even when its public column list remains unchanged.

Session data is operational

The sessions table contains authentication state, not production-domain records. It is isolated from all business-data joins.

MAINTENANCE Regenerate this document whenever a source object, join key, selected column, transformation, or session-storage rule changes.